

Biomass CHP — Maker Shortlist

Potential manufacturers for the Mitsue biomass CHP, split by the two build stages:

Tier A — a 10–20 kWe first prototype, and **Tier B — the 2 × 0.6 MWe (1.2 MWe) commercial fleet**.

Policy reminder: the *commercial* units must be a **Japanese manufacturer** (domestic service/parts, simpler FIT + 交付金 paperwork, local supply-chain story). Imported units are **benchmark-only, not for procurement** — with one possible, explicitly-flagged exception at prototype scale (see Tier A).

Authoritative reference: the JWBA list 国内で販売されている小規模木質バイオマス発電機器の一覧 catalogues every domestic unit with specs — use it to confirm any figure below.

⚠ All kW ratings here are **indicative / to verify** against the JWBA list or a vendor quote before use. CHP sizing in the wider project is still a planning estimate.

The key finding — the prototype scale has no domestic gasifier

Domestic woody-**gasification** CHP has a floor around ~**200–350 kWe** (Neonite's smallest is ~350 kWe; Shizuoka Seiki ~200 kWe). A 10–20 kWe prototype is an **order of magnitude below that floor**, so there is **no domestic-maker, gasification unit at prototype scale**.

This forces an explicit choice for the prototype (a decision, not a spec gap):

- **(a) Domestic non-gasification micro-CHP** — a wood-chip **boiler + Stirling engine** at ~10 kWe *does* exist domestically (巴商会). Keeps the Japanese-maker policy intact; trades gasification for Stirling/combustion at this stage.
- **(b) Imported micro-gasifier as a one-off pilot exception** — All Power Labs / Volter / Spanner HKA-class. Matches the eventual gasification tech, but breaks the domestic-maker policy for the prototype only — would need to be justified as pilot-only.
- **(c) Research / university partnership build** — slower, but keeps IP and learning local.

Recommendation to decide: **(a) for a fast, policy-clean, revenue-visible prototype**, or **(b) if the prototype's main job is to de-risk the exact gasification tech** we scale to.

Tier A — 10–20 kWe first prototype

Maker	Tech / scale	Domestic?	Notes
巴商会 (Tomoe Shokai)	wood-chip boiler + Stirling engine , ~10 kWe / ~180 kWth	 (distributor)	Closest domestic match to prototype scale; combustion+Stirling, <i>not</i> gasification. Lead for option (a).

Maker	Tech / scale	Domestic?	Notes
ネオナイト (Neonite)	downdraft 2-stage gasification , smallest ~350 kWe	✓	Too big for a prototype — belongs to Tier B. Biochar/wood-tar recovery.
静岡製機 (Shizuoka Seiki)	small outdoor gasification, ~200 kWe	✓	Also above prototype scale; Tier B cluster candidate.
All Power Labs (Power Pallet)	downdraft micro- gasification , ~10–25 kWe	✗ US (on JWBA list)	Best <i>gasification</i> size match; import → pilot-exception only .
Volter	CHP micro-gasifier, ~40 kWe	✗ Finland	Benchmark; import-only.
Spanner Re ² (HKA-9 / HKA-30 / HKA-45)	wood gasifier-CHP modules, ~9–45 kWe	✗ Germany	Benchmark; smallest module near prototype scale; import-only.

Tier B — 2 × 0.6 MWe (1.2 MWe) commercial fleet

Small woody-gasification tops out ~200–500 kWe, so 0.6 MWe per line is a **cluster of modules**. (Reproduced from [mitsue_forest_workforce_energy_plan.md](#) §5.)

Maker	Tech / scale	Fit	Notes
中外炉工業 (Chugai Ro)	rotary-kiln gasification cogen	✓ lead	Since 1997, NEDO-backed; reaches our per-unit size; tolerates rough fuel (branches/bark).
神鋼環境ソリューション (Shinko / Kobe Steel grp)	~500 kWe-class up	backup	⚠ confirm gasification vs steam (big plants are combustion).
静岡製機 (Shizuoka Seiki)	small ~200 kWe outdoor unit	cluster	METI-funded, aimed at agriculture/forestry — closest in spirit.
ネオナイト (Neonite)	downdraft two-stage, ~350 / 750 kWe	cluster	Recovers charcoal + wood tar (biochar angle).

Imported — benchmark-only, not for procurement: FORTES Energy (Latvia; 0.14–1.2 MWe gasification CHP, new Tokyo entity), Spanner Re² and Burkhardt (Germany; e.g. V3.90 module). Useful for spec comparison; do not meet the Japanese-manufacturer requirement.

Next steps

- Pull the current **JWBA small-scale list** and confirm every kW figure above.

- Decide the prototype path — option (a) / (b) / (c) — before requesting quotes.
- Vendor/site-visit outreach already drafted in `mitsue_biomass_visit_request_emails.md` (中外炉 / 神鋼 / 静岡製機 / ネオナイト priority; Spanner / Burkhardt as imported benchmarks).

Sources

- JWBA 小規模木質バイオマス発電機器の一覧 — <https://jwba.or.jp/database/list-small-woody-biomass-generation/>
- JWBA list PDF (alphabetical, incl. All Power Labs) — https://jwba.or.jp/wp/wp-content/uploads/2024/11/20241122_国内で販売されている小規模木質バイオマス発電機器の一覧.pdf
- 巴商会 木質バイオマスボイラ + 発電 (Stirling, ~10 kWe/180 kWth) — https://tomoeshokai.com/product/woody_biomass_boiler/generator
- ネオナイト 小規模木質バイオマスガス化発電 (350/750 kW) — http://www.neonite.jp/biomass_lp/
- Metoree バイオマス発電機メーカー一覧 — <https://metoree.com/categories/2730/>

Related docs

- `mitsue_forest_workforce_energy_plan.md` §5 (energy stack) — CHP maker table source
- `mitsue_biomass_visit_request_emails.md` — vendor visit outreach